STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 9: Mechanical Properties of Fluids

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

1. Define pressure. [1]
2. SI unit of dynamic viscosity? [1]
3. Continuity relation for incompressible flow? [1]
4. Bernoulli relation? [1]
5. Surface tension unit? [1]
6. Pressure rise at depth h? [1]
7. Stokes drag? [1]
8. Excess pressure in a liquid drop? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

9. Water at 3m depth, rho=1000,g=10: gauge pressure? [2]
10. If pipe area halves, incompressible speed? [2]
11. Under Bernoulli assumptions, faster horizontal flow means pressure? [2]
12. Force from p=5000 Pa over area .02 m^2? [2]
13. Drop S=.06 N/m,r=.003 m: excess pressure? [2]
14. Why is a soap bubble pressure difference twice a drop's? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

15. Pipe area 4 cm^2 at 2 m/s narrows to 1 cm^2. New speed? [3]
16. Horizontal water speeds 2 and 6 m/s, rho=1000. Find p_1-p_2. [3]
17. Stokes drag for r=.5 mm,=1 Pa s,v=.02 m/s? [3]
18. Explain why pressure rises with depth. [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

19. Areas 6 cm^2 and 2 cm^2; inlet speed 1.5 m/s. Find outlet speed. [4]
20. Soap bubble r=2mm,S=.03N/m: excess pressure? [4]

End of paper - check working, units, signs and significant figures.